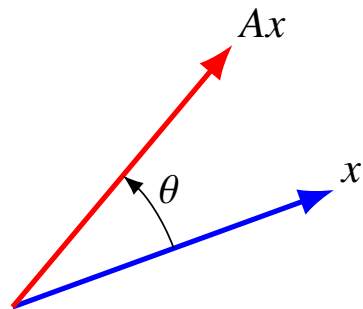


Geometric transformations

- ▶ many geometric transformations and mappings of 2-D and 3-D vectors can be represented via matrix multiplication $y = Ax$
- ▶ for example, rotation by θ :

$$y = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} x$$



(to get the entries, look at Ae_1 and Ae_2)

Outline

Geometric transformations

Selectors

Incidence matrix

Convolution

Selectors

- ▶ an $m \times n$ selector matrix: each row is a unit vector (transposed)

$$A = \begin{bmatrix} e_{k_1}^T \\ \vdots \\ e_{k_m}^T \end{bmatrix}$$

- ▶ multiplying by A selects entries of x :

$$Ax = (x_{k_1}, x_{k_2}, \dots, x_{k_m})$$

- ▶ example: the $m \times 2m$ matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & 0 \end{bmatrix}$$

‘down-samples’ by 2: if x is a $2m$ -vector then

$$y = Ax = (x_1, x_3, \dots, x_{2m-1})$$

- ▶ other examples: image cropping, permutation, ...

8. Linear equations

Outline

Linear functions

Linear function models

Linear equations

Balancing chemical equations

Superposition

- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ means f is a function that maps n -vectors to m -vectors
- ▶ we write $f(x) = (f_1(x), \dots, f_m(x))$ to emphasize components of $f(x)$
- ▶ we write $f(x) = f(x_1, \dots, x_n)$ to emphasize components of x
- ▶ f satisfies *superposition* if for all x, y, α, β

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

(this innocent looking equation says a lot ...)

- ▶ such an f is called *linear*

Matrix-vector product function

- ▶ with A an $m \times n$ matrix, define f as $f(x) = Ax$
- ▶ f is linear:

$$\begin{aligned} f(\alpha x + \beta y) &= A(\alpha x + \beta y) \\ &= A(\alpha x) + A(\beta y) \\ &= \alpha(Ax) + \beta(Ay) \\ &= \alpha f(x) + \beta f(y) \end{aligned}$$

- ▶ converse is true: if $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is linear, then

$$\begin{aligned} f(x) &= f(x_1 e_1 + x_2 e_2 + \cdots + x_n e_n) \\ &= x_1 f(e_1) + x_2 f(e_2) + \cdots + x_n f(e_n) \\ &= Ax \end{aligned}$$

$$\text{with } A = \begin{bmatrix} f(e_1) & f(e_2) & \cdots & f(e_n) \end{bmatrix}$$

Examples

- reversal: $f(x) = (x_n, x_{n-1}, \dots, x_1)$

$$A = \begin{bmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{bmatrix}$$

- running sum: $f(x) = (x_1, x_1 + x_2, x_1 + x_2 + x_3, \dots, x_1 + x_2 + \cdots + x_n)$

$$A = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & 1 & 0 \\ 1 & 1 & \cdots & 1 & 1 \end{bmatrix}$$

Affine functions

- ▶ function $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is *affine* if it is a linear function plus a constant, i.e.,

$$f(x) = Ax + b$$

- ▶ same as:

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

holds for all x, y , and α, β with $\alpha + \beta = 1$

- ▶ can recover A and b from f using

$$\begin{aligned} A &= \begin{bmatrix} f(e_1) - f(0) & f(e_2) - f(0) & \cdots & f(e_n) - f(0) \end{bmatrix} \\ b &= f(0) \end{aligned}$$

- ▶ affine functions sometimes (incorrectly) called linear

Outline

Linear functions

Linear function models

Linear equations

Balancing chemical equations

Linear and affine functions models

- ▶ in many applications, relations between n -vectors and m -vectors are *approximated* as linear or affine
- ▶ sometimes the approximation is excellent, and holds over large ranges of the variables (e.g., electromagnetics)
- ▶ sometimes the approximation is reasonably good over smaller ranges (e.g., aircraft dynamics)
- ▶ in other cases it is quite approximate, but still useful (e.g., econometric models)

Regression model

- ▶ regression model: $\hat{y} = x^T \beta + v$
 - x is n -vector of features or regressors
 - β is n -vector of model parameters; v is offset parameter
 - (scalar) $\hat{y}$ is our prediction of y
- ▶ now suppose we have N *examples* or *samples* $x^{(1)}, \dots, x^{(N)}$, and associated responses $y^{(1)}, \dots, y^{(N)}$
- ▶ associated predictions are $\hat{y}^{(i)} = (x^{(i)})^T \beta + v$
- ▶ write as $\hat{\mathbf{y}}^d = X^T \beta + v \mathbf{1}$
 - X is feature matrix with columns $x^{(1)}, \dots, x^{(N)}$
 - $\mathbf{y}^d$ is N -vector of responses $(y^{(1)}, \dots, y^{(N)})$
 - $\hat{\mathbf{y}}^d$ is N -vector of predictions $(\hat{y}^{(1)}, \dots, \hat{y}^{(N)})$
- ▶ *prediction error* (vector) is $\mathbf{y}^d - \hat{\mathbf{y}}^d = \mathbf{y}^d - X^T \beta - v \mathbf{1}$

Outline

Linear functions

Linear function models

Linear equations

Balancing chemical equations

Systems of linear equations

- ▶ set (or *system*) of m linear equations in n variables $x_1, \dots, x_n$:

$$\begin{aligned}A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= b_1 \\A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= b_2 \\&\vdots \\A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= b_m\end{aligned}$$

- ▶ n -vector x is called the variable or unknowns
- ▶ A_{ij} are the *coefficients*; A is the coefficient matrix
- ▶ b is called the *right-hand side*
- ▶ can express very compactly as $Ax = b$

Systems of linear equations

- ▶ systems of linear equations classified as
 - under-determined if $m < n$ (A wide)
 - square if $m = n$ (A square)
 - over-determined if $m > n$ (A tall)
- ▶ x is called a *solution* if $Ax = b$
- ▶ depending on A and b , there can be
 - no solution
 - one solution
 - many solutions
- ▶ we'll see how to solve linear equations later